IMAGE II TABLE OF CONTENTS

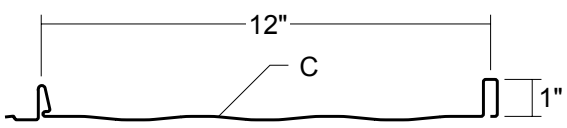
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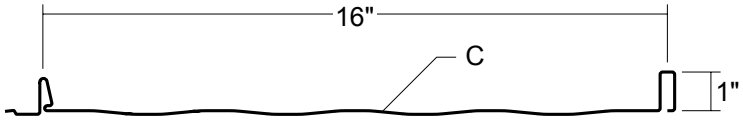
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IMAGE II Panel Overview

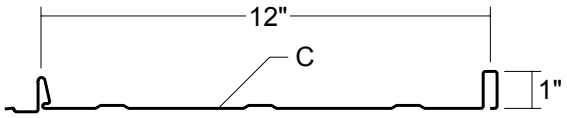
PANEL PROFILES



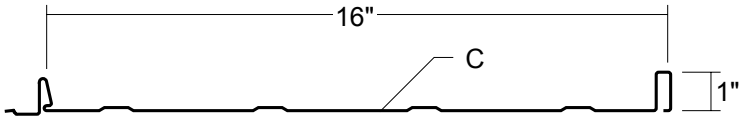
12" Coverage
Striations



16" Coverage
Striations



12" Coverage
Minor Ribs



16" Coverage
Minor Ribs

SLOPE

The minimum recommended slope for any Image II roofing panel is 3:12.

SHEATHING

Plywood and OSB are the typical sheathing materials. Common plywood thicknesses include 1/2" and 5/8". Common OSB thicknesses include 1/2" and 3/4". To avoid panel distortion, use a properly aligned and uniform substructure. **Please note that Image II panels are not recommended for use over open framing.**

UNDERLAYMENT

An underlayment is recommended and required by building codes. 30# felt will meet the minimum performance level. Synthetic underlayments provide improved performance and peel-and-stick underlayments provide further improved performance.

COVERAGE

Image II panels are available in a 12" and 16" widths with a 1" rib height.

LENGTH

Minimum panel length is 5'-0". Maximum recommended panel length is 30'-0". Longer panels require additional consideration in packaging, shipping, and erection. Please consult your Metal Sales branch for recommendations (see PGI-2 and 3 for locations).

AVAILABILITY

Image II panels material gauges and finishes vary. Please contact your local branch for options and availability.

APPLICATION

Applications include residential, architectural and commercial buildings.

PERFORMANCE TEST

UL 2218, UL790, UL263, ASTM E-1592

FASTENERS

The fastener selection guide should be consulted for choosing proper fasteners for specific applications. Quantity and type of fastener must meet necessary loading and code requirements (see PGI-12-13).

MATERIALS

26 and 24 gauge steel: Grade 50 per ASTM A 792

FINISH

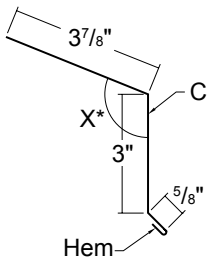
- *Acrylic Coated Galvalume® (ACG) / ASTM A 792
- Prepainted Galvalume® (ACG), AZ50 per ASTM A 792
- MS Colorfast45®
- **PVDF

* Differential appearance of Acrylic Coated Galvalume roofing materials is not a cause for rejection.
** Meets both Kynar 500 and Hylar 5000 specifications.

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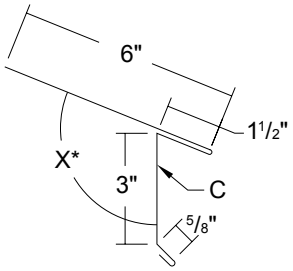
IMAGE II FLASHING PROFILES

EAVE



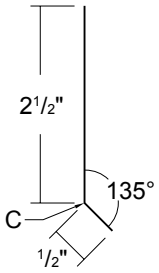
Length 10'-2" - *Specify Slope Angle

EXTENDED EAVE



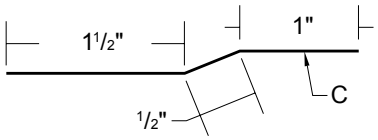
Length 10'-2" - *Specify Slope Angle

CLEAT



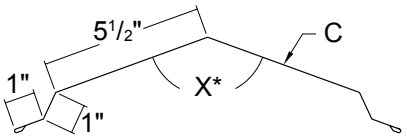
Length 10'-2"

OFFSET CLEAT



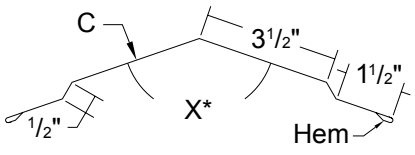
Length 10'-2"

13" RIDGE/HIP COVER



Length 10'-2", 20'-3" - *Specify Slope Angle

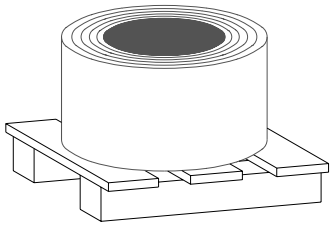
STEP RIDGE/HIP COVER



Length 10'-2" - *Specify Slope Angle

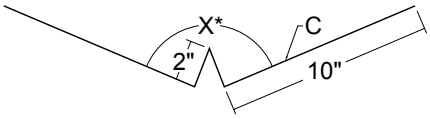
12" COIL

(For Continuous Gutters)



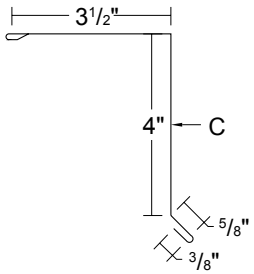
16" Inside Diameter
Length 150'-0"

VALLEY



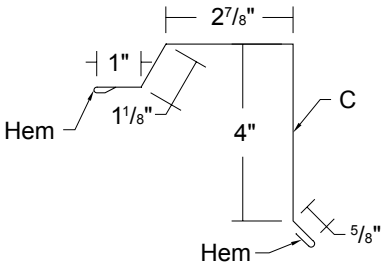
Length 10'-2", 20'-3" - *Specify Slope Angle

IMAGE II RAKE



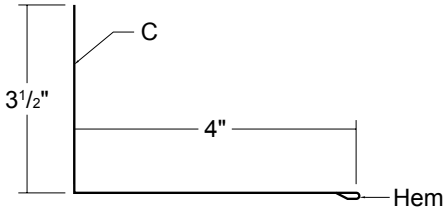
Length 10'-2", 20'-3"

IMAGE II STEP RAKE



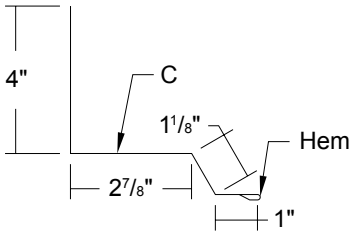
Length 10'-2", 20'-3"

RAKEWALL



Length 10'-2"

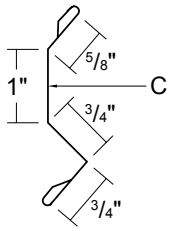
IMAGE II STEP RAKEWALL



Length 10'-2"

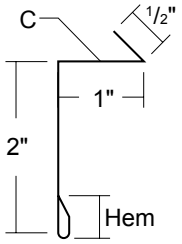
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COUNTER FLASHING



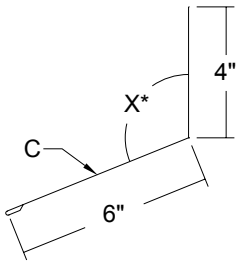
Length 10'-2"

REGLET FLASHING



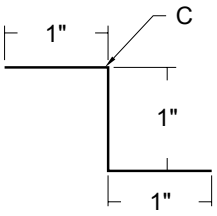
Length 10'-2"

PITCH BREAK



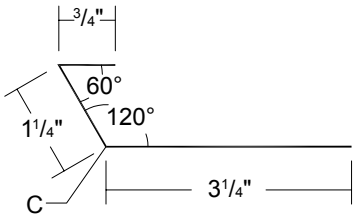
*Length 10'-2" - *Specify Slope Angle*

1" Z-CLOSURE



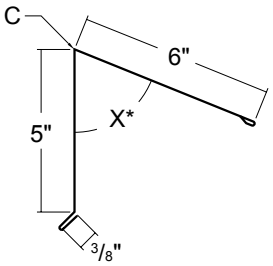
Length 10'-2"

PERFORATED VENT DRIP



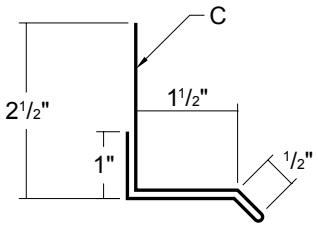
.032 Aluminum
Linen White Only
Length 10'-2"

PEAK



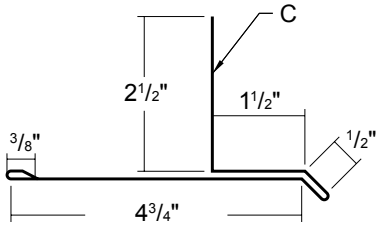
*Length 10'-2", 20'-3" - *Specify Slope Angle*

1.5" SILL/HEAD



Length 10'-2"

1.5" SILL TO SOFFIT

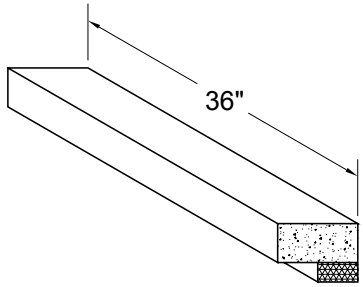


Length 10'-2"

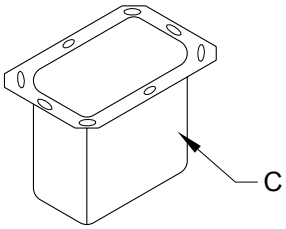
C- Indicates color side of flashing.

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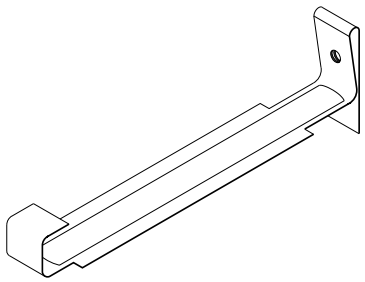
COBRA VENTED CLOSURE



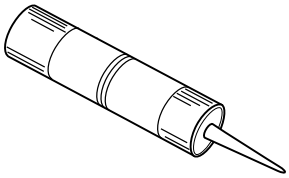
OUTLET TUBE 2" X 3"



FACIA BRACKET



TUBE SEALANT



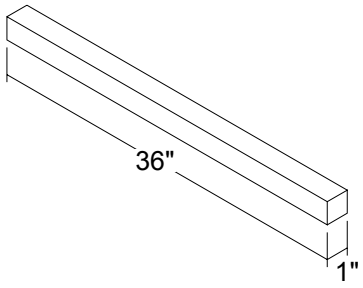
10.3 oz. Cartridge
Urethane

TAPE SEALANT

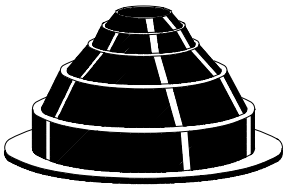


$^{7}/_{8}$" X $^{3}/_{16}$" X 25'
Double Bead
Butyl - Gray

OUTSIDE CLOSURE

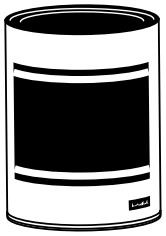


RUBBER ROOF JACK



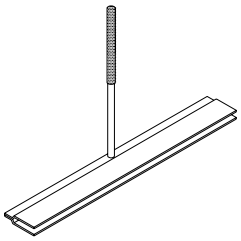
MINI ($^{1}/_{4}$" to $1^{1}/_{8}$" O.D. Pipe)
#2 ($1^{3}/_{4}$" to 3" O.D. Pipe)
#4 (3" to 6" O.D. Pipe)
#6 (6" to 9" O.D. Pipe)
#8 (7" to 13" O.D. Pipe)

TOUCH-UP PAINT



Available in pints
PVDF / MS Colorfast45

METAL PANEL HEMMING TOOL



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Metal Roof Deck Panels

Metal Sales Manufacturing Corporation has obtained fire resistance ratings for various products conducted according to test criteria set forth by 'Underwriters Laboratories' "Standard Fire Tests of Building Construction and Material" (ANSI/UL 263). This test procedure is identical to ASTM E-119 and NFPA 251.

The fire resistance rating is for the total assembly and not just the external metal panel. Ratings are expressed in hours and vary depending upon the assemblies.
In general, the test criteria is to evaluate the assembly's ability to continue to support the super-imposed loads and resist the passage of flame, high temperatures, or hot gases which will ignite combustible materials. The test assemblies are identified by an alpha-numeric design number.

For detail information on specific assemblies and hourly ratings see UL Fire Resistance Directory.

METAL SALES MFG CORP
R9697

Mechanically attached metal roof panels - Type “Image II”

For use in Design Nos. P224 , P225 , P227 , P230 , P237 , P508 , P510 , P512 , P701 , P711 , P712 , P713 , P715 , P717 , P720 , P722 , P723 , P724 , P726 , P731 , P734 , P736 , P803 , P814 , P815 , P818 , P819 , P821 , P823 , P824 .

*Hat shaped member to be a minimum of 16 gauge. The member will be fastened through the roof insulation to the steel roof deck with min. No. 14 self-drilling and/or self-tapping fasteners. Spacing to be determined by the structural loading requirements. In addition any compressible UL Classified glass fiber blanket insulation with or without a vapor retarder facing may be used between the specified roof insulation and the metal roof panels.

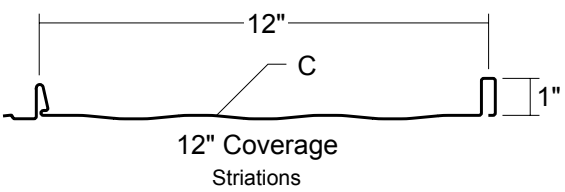
**Bearing plate to be a minimum of 16 gauge. Member will be fastened through the roof insulation to the steel deck with min. No. 14 self-drilling and/or self-tapping fasteners.

See the UL Fire Resistance Directory for explanation of each design number listed above.

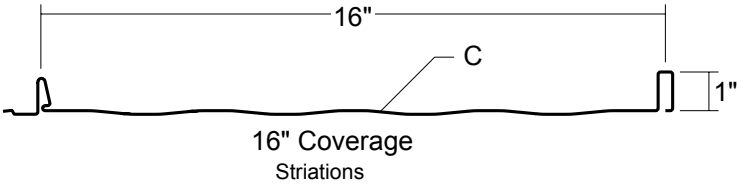
Underwriters Laboratories Inc. ®

LISTED

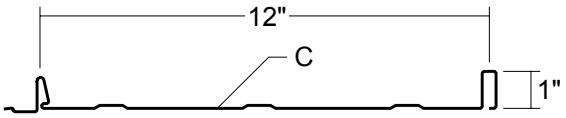
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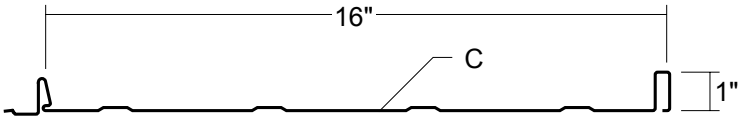
12" Coverage
Striations



16" Coverage
Striations



12" Coverage
Minor Ribs

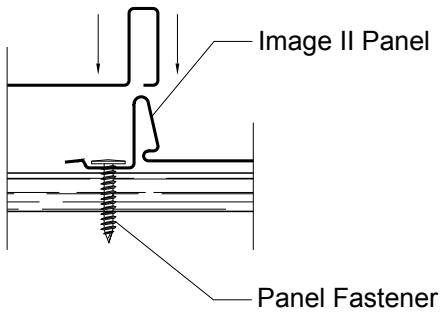


16" Coverage
Minor Ribs

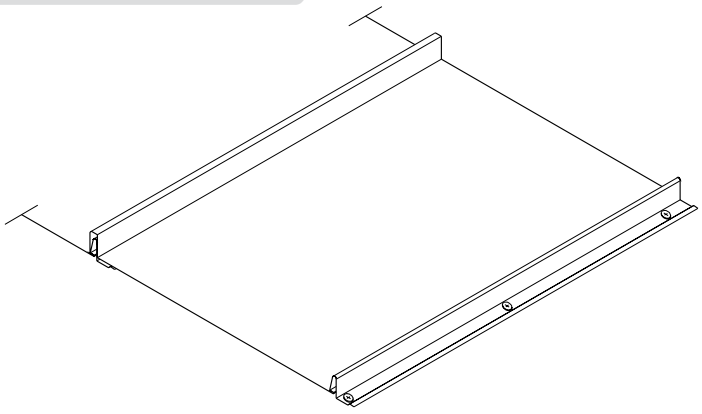
SECTION PROPERTIES								ALLOWABLE UNIFORM LOADS, psf For various fastener spacings			
Ga	Width in	Yield ksi	Weight psf	Top In Compression		Bottom In Compression		Outward Load			
				Ixx in^4/ft	Sxx in^3/ft	Ixx in^4/ft	Sxx in^3/ft	0.5'	1'	1.5'	2'
26	12	50	0.99	0.0209	0.0231	0.0211	0.0232	103	96	90	84
24	12	50	1.28	0.0268	0.0295	0.0268	0.0295	103	96	90	84
26	16	50	0.92	0.0165	0.0174	0.0165	0.0177	103	96	90	84
24	16	50	1.19	0.0210	0.0226	0.0210	0.0226	103	96	90	84

1. Theoretical section properties have been calculated per AISI 2016 'North American Specification for the Design of Cold-Formed Steel Structural Members'. Ixx and Sxx are effective section properties for deflection and bending.
2. Allowable load is calculated in accordance with AISI 2016 specifications considering bending, shear, combined bending & shear and deflection. Allowable load does not address web crippling, fasteners, support material or load testing. Allowable load considers the three or more equal spans condition. Panel weight is not considered.
3. Deflection consideration is limited by a maximum deflection ratio of L/180 of span.
4. Allowable loads do not include a 1/3 stress increase for wind.

ATTACHMENT DETAIL



FASTENING PATTERN



FASTENER INFORMATION

Overdriven fasteners will cause panel distortions.

Fasteners should extend 1/2" or more past the inside face of the support material.

Thick panels (ex. 18 ga) or supports (ex. 1/2" steel) may require predrilling of holes for screws.

Panel Fastener:
#10-12 Pancake Head Wood Screw
or
#8-15 Truss Head Wood Screw

Concealed End Fastener:
#10-12 Pancake Head Wood Screw
or
#8-15 Truss Head Wood Screw

Exposed End Fastener:
#10-14 XL Wood Screw

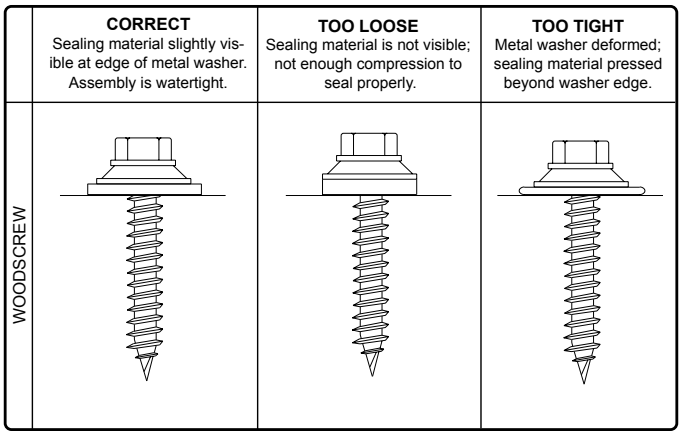
Trim Fastener:
1/4"-14 x 7/8" XL Stitch Screw
or
1/8" x 3/16" Pop Rivet

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FASTENER INSTALLATION TECHNIQUE

Recommended Tool Type - Use depth locating nose or adjustable clutch on screw gun to prevent overdrilling and strip out. **Do not use impact tools or runners.**

Seating the washer - Apply sufficient torque to seat the washer - do not overdrive the fastener.



To prevent wobbling - Make sure fastener head is completely engaged in the socket. If the head does not go all the way in the socket - tap the magnet deeper into the socket to allow full head engagement. Metal chips will build up from drilling and should be removed from time to time.

Protect drill point - Push only hard enough on the screw gun to engage clutch. This prevents excess friction and burn out of the drill point. Correct pressure will allow screw to drill and tap without binding.

Drilling through sheet and insulation - Ease up on pressure when drilling through insulation to avoid striking the purlin or girt with the point - apply more pressure after drill point contacts purlin or girt.

Drilling through purlin overlaps - Drilling through lapped purlins requires extra care. Excessive voids between purlins sometimes damages drill points and two self-drillers might be necessary to complete the operation. It is sometimes advantageous to predrill.

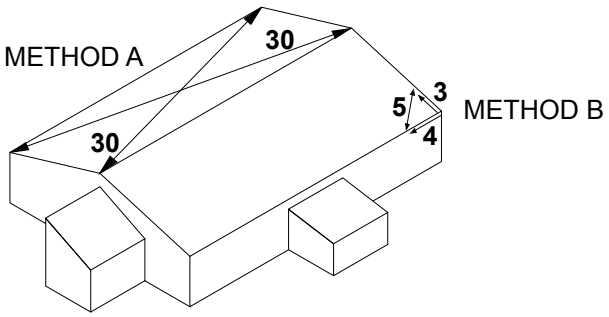
CONDITION OF SUBSTRUCTURE

Whether over solid substrate or open structural framing, panel distortion may occur if not applied over properly aligned and uniform substructure.

The installer should check the roof deck for squareness before installing Image II panels. Several methods can be used to verify squareness of the structure for proper installation of the panels.

METHOD "A" - One method for checking the roof for squareness is to measure diagonally across one slope of the roof from similar points at the ridge and eave and obtain the same dimension.

METHOD "B" - The 3-4-5 triangle system may also be used. To use this system measure a point from the corner along the edge of the roof at a module of three (3). Measure a point from the same corner along another edge at a module of four (4). Then by measuring diagonally between the two points established, the dimension should be exactly a module of five (5) to have a square corner. Multiple uses of this system may be required to determine building squareness. If the endwall cannot be made square, the roof system cannot be installed as shown in these instructions.



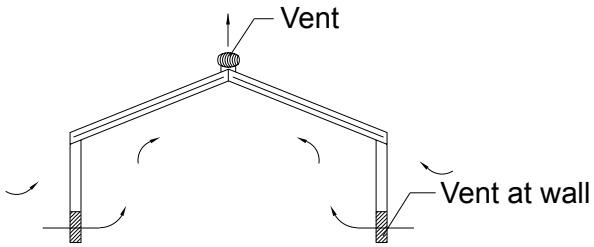
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VENTILATION

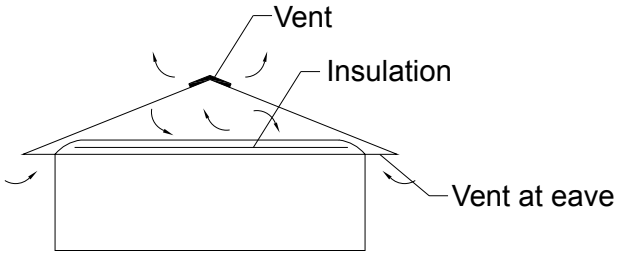
Proper design and installation of vapor barriers and ventilation systems are important to prevent condensation and the resulting problems of moisture damage and loss of insulation efficiency.

Condensation occurs when moisture laden air comes in contact with a surface temperature equal to or below the dew point of the air. This phenomenon creates problems that are not unique with metal roofing; these problems are common to all types of construction.

The underside of the metal roof on a typical Architectural building should be protected from condensation by installing panels directly over a minimum 30 lb moisture barrier and uniform solid substrate. This reduces airspace and the potential of condensation forming on the underside of the panels.



Typical metal building (no attic)



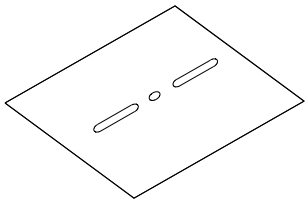
Building with attic or retrofitted

PANEL APPLICATIONS

The following chart highlights UL 580 Class 90 for clip installation on the selected applications (see Fastener Selection Guide page PGI-12-14 for other fasteners available). For more information on UL Construction numbers, refer to UL Roofing Materials and System Directories. Panel gauges and clip spacing should be determined by a professional engineer according to the governing building code.

PANEL TYPE	APPLICATION	INSTALLATION REQUIREMENTS		FASTENER SPACING	*TYPE OF FASTENER	NUMBER REQUIRED
IMAGE II	OVER 5/8" WOOD DECK AND METAL DECK	Standard	26 GAUGE	18" o.c.	CALL YOUR METAL SALES BRANCH FOR ASTM-E1592 UPLIFT VALUE	

***** Subject to project loading, closer fastener spacing may be required. Contact your local Metal Sales branch representative for more information (see pages PGI-2-3). Fastener spacing is based on maximum uplift suction load of 32 psf.

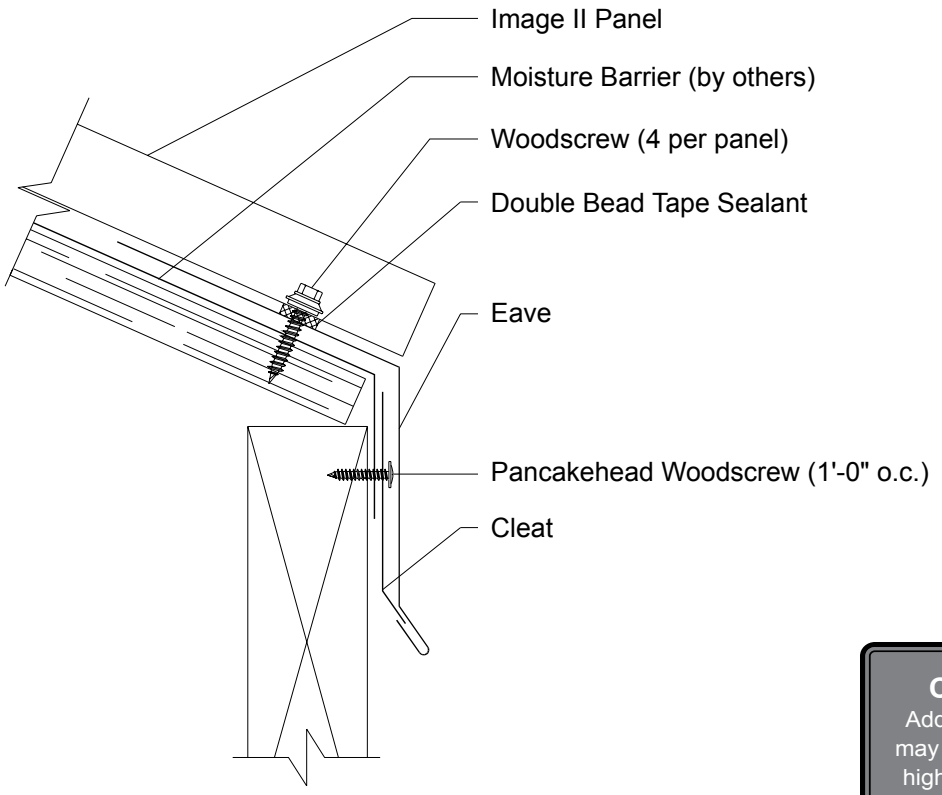


BEARING PLATE

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IMAGE II EAVE DETAIL

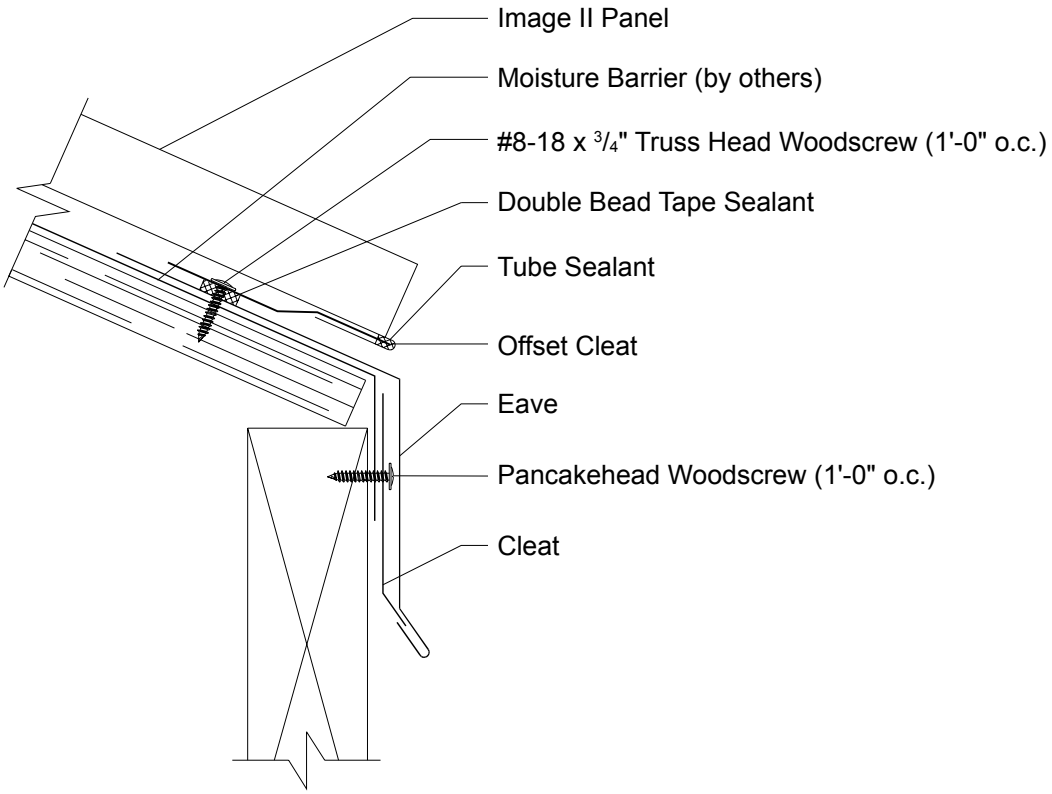
3:12 Slope Minimum



CAUTION
Additional screws may be required for high snow loading and steep slopes.

IMAGE II EAVE WITH OFFSET DETAIL

3:12 Slope Minimum

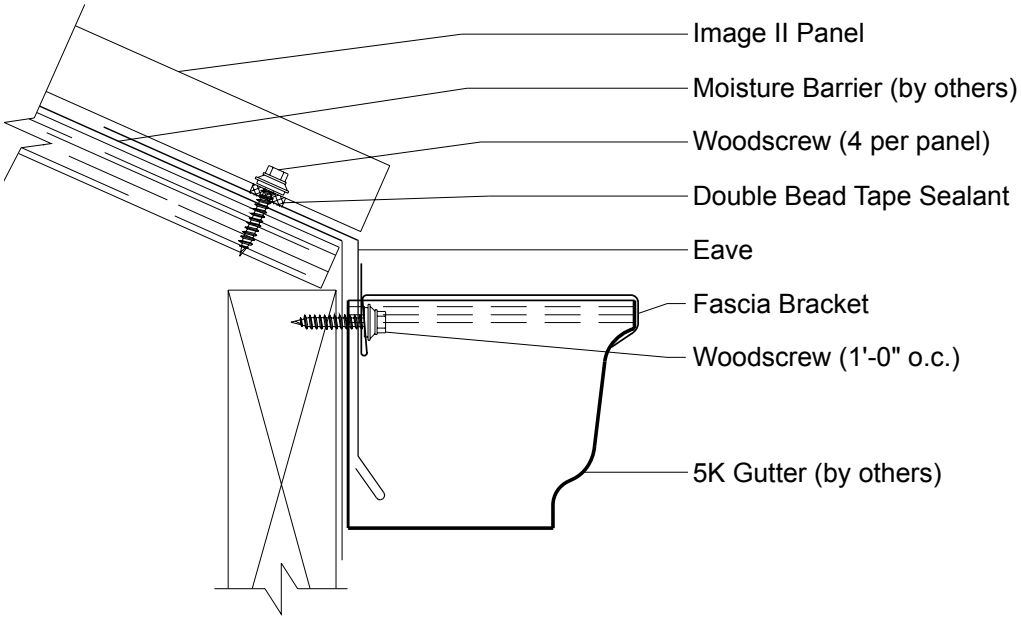


Panel rib must be field notched and flat part of panel must be field bent to accept offset cleat (see Architectural Manual).

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IMAGE II 5K Gutter Detail

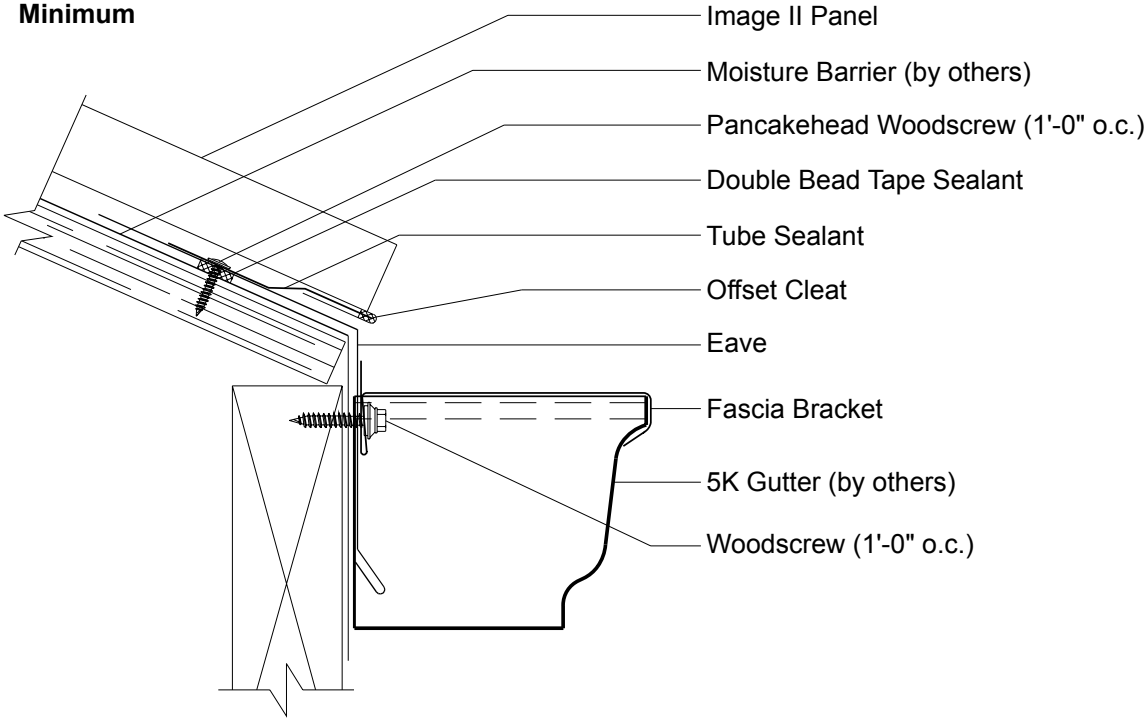
3:12 Slope Minimum



CAUTION
Additional screws may be required for high snow loading and steep slopes.

IMAGE II 5K Gutter with Offset Detail

3:12 Slope Minimum

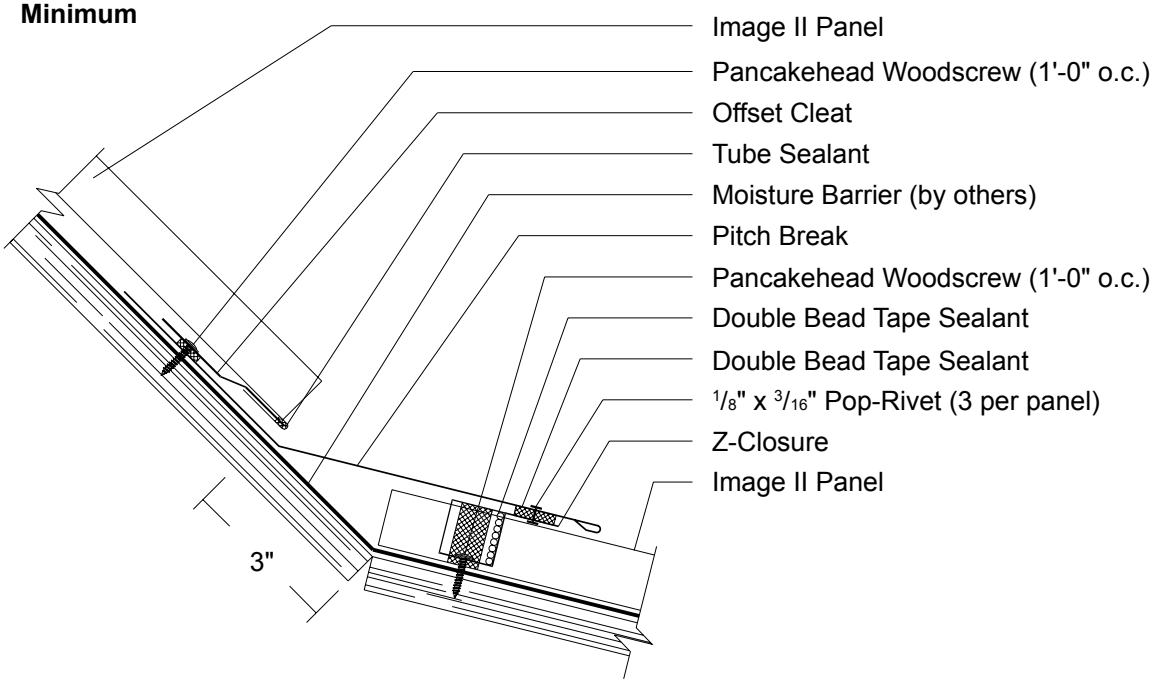


Panel rib must be field notched and flat part of panel must be field bent to accept offset cleat (see Architectural Manual).

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IMAGE II SLOPE CHANGE DETAIL

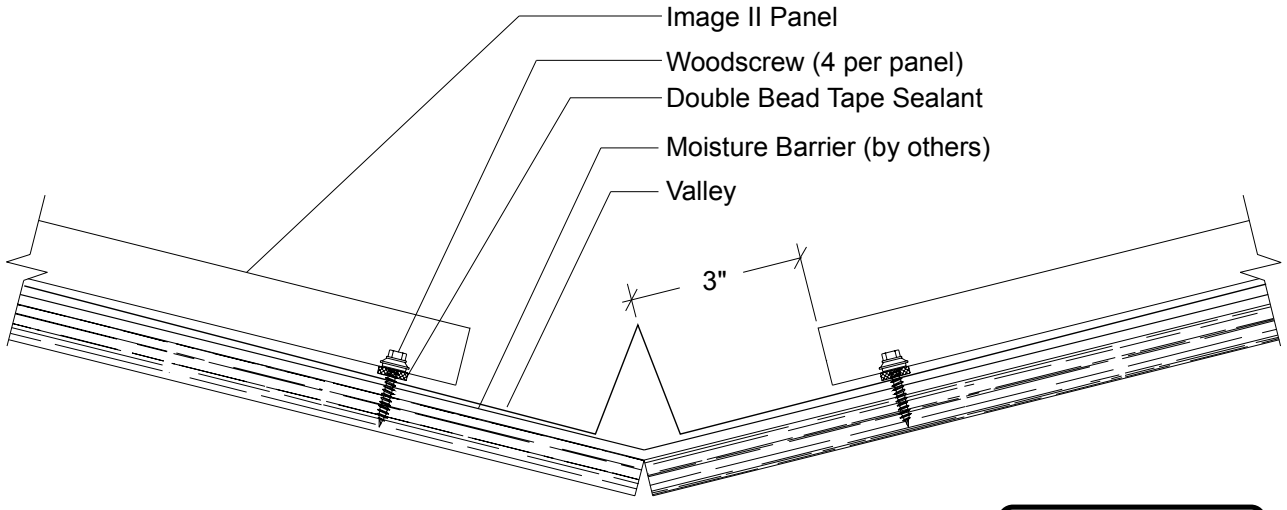
3:12 Slope Minimum



Note: Z-Closures must be field cut and bent to fit between panel ribs (see Architectural Manual).
Panel rib must be field notched and flat part of panel must be field bent to accept offset cleat (see Architectural Manual).

IMAGE II VALLEY DETAIL

3:12 Slope Minimum

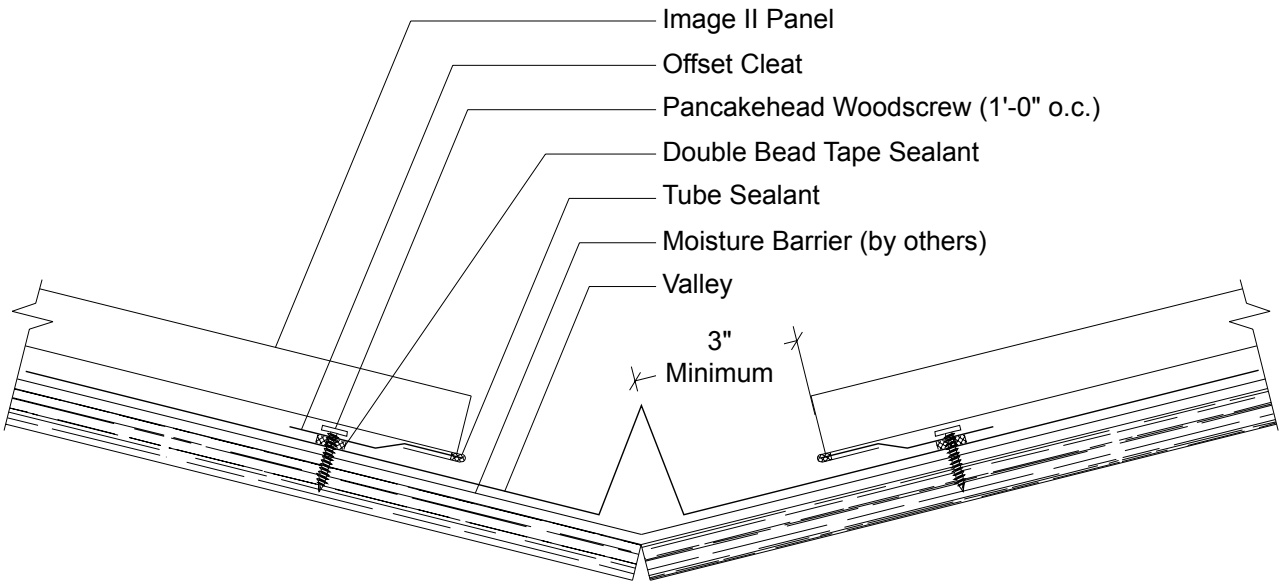


CAUTION
Additional screws may be required for high snow loading and steep slopes.

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IMAGE II Valley with Offset Cleat Detail

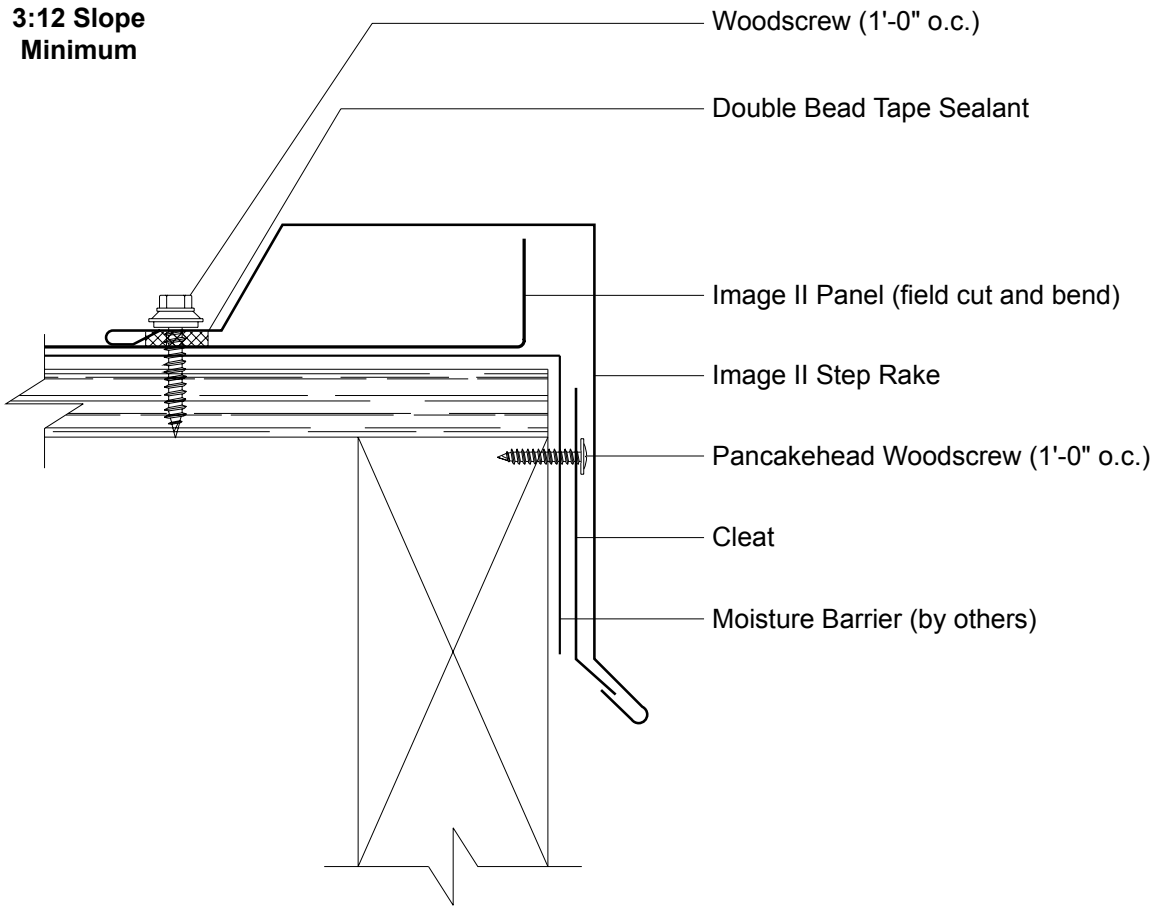
3:12 Slope Minimum



Panel rib must be field notched and flat part of panel must be field bent to accept offset cleat (see Architectural Manual).

IMAGE II Step Rake Detail

3:12 Slope Minimum



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IMAGE II RAKE WITH Z-CLOSURE DETAIL

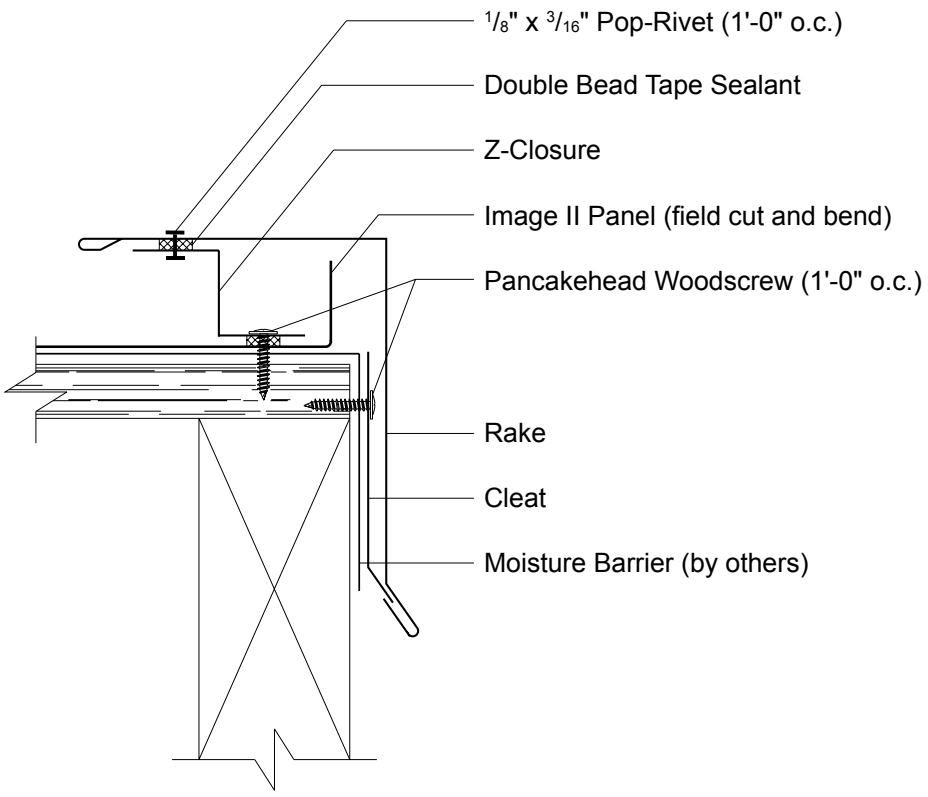
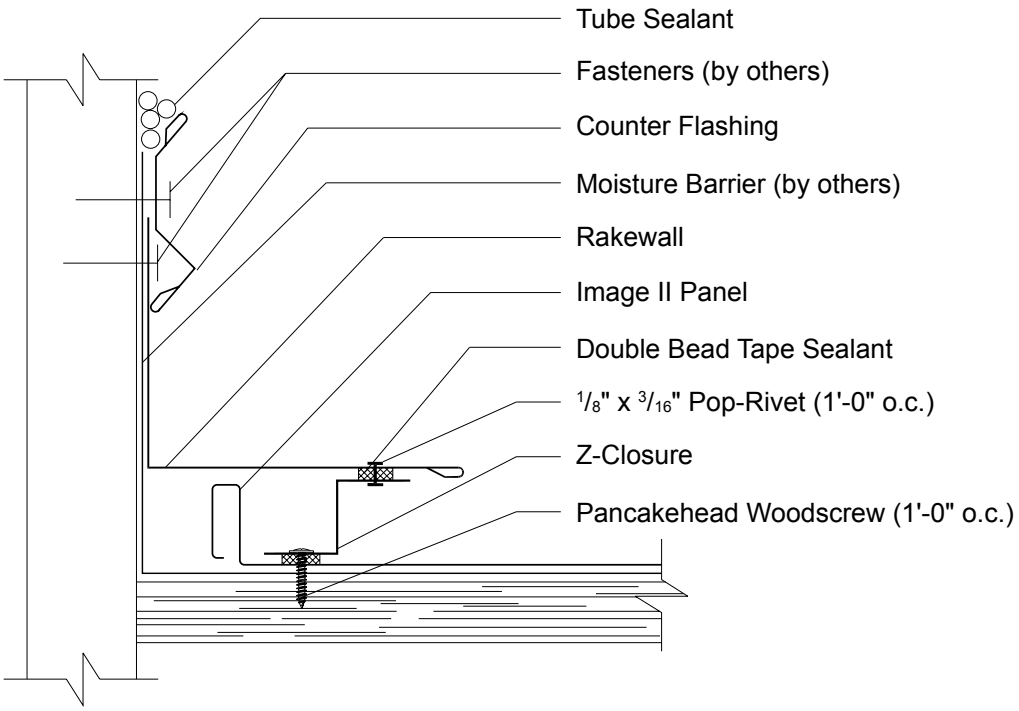


IMAGE II RAKEWALL WITH Z-CLOSURE (COUNTER) DETAIL



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IMAGE II Rakewall with Z-Closure (Reglet) Detail

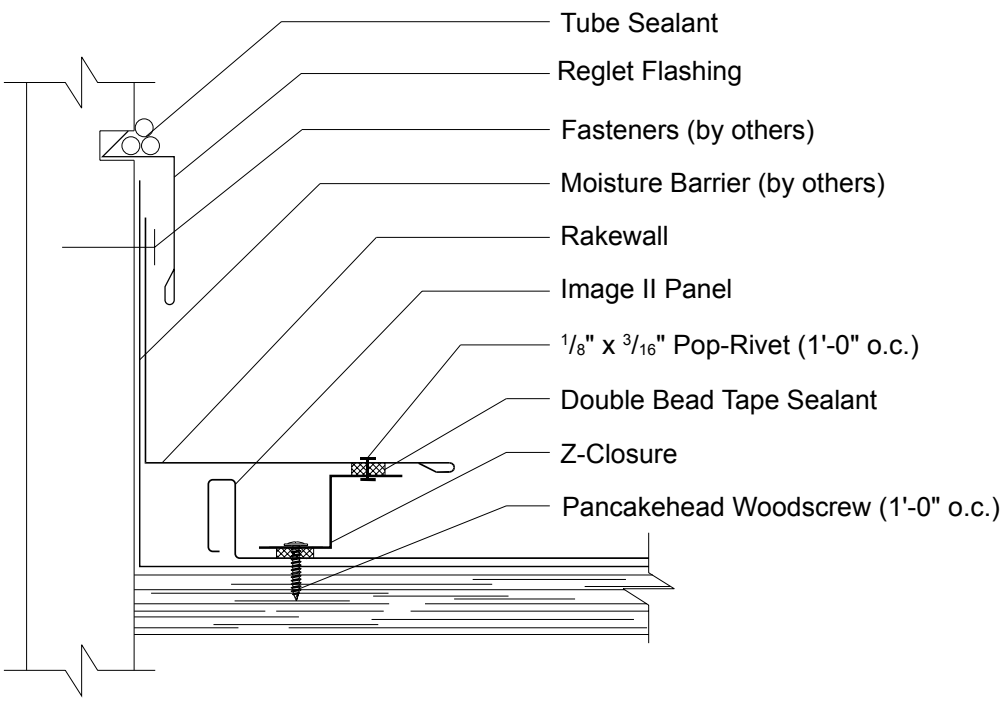
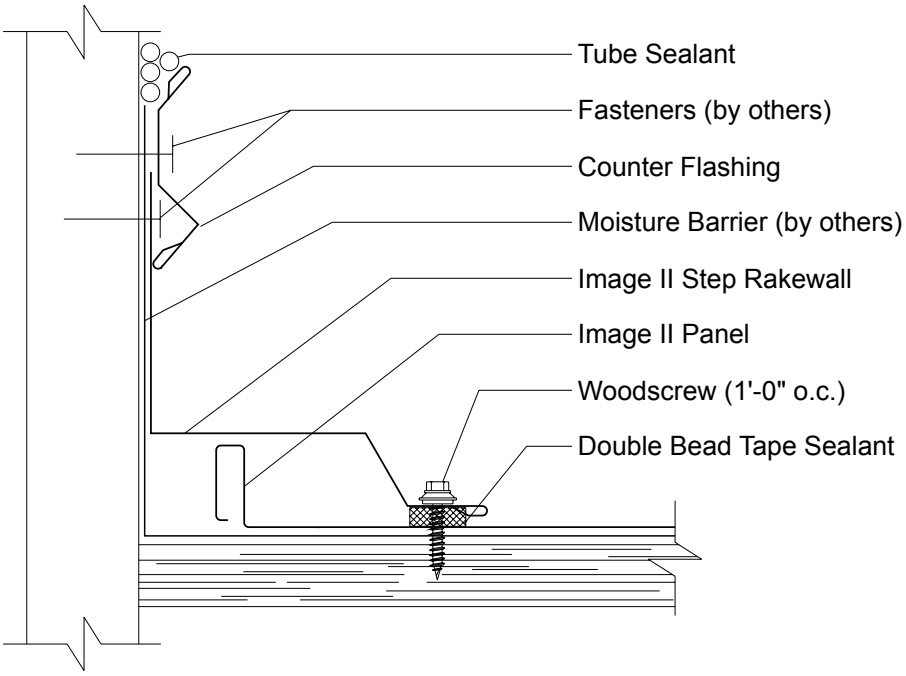


IMAGE II Step Rakewall (Counter) Detail



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IMAGE II STEP RAKEWALL (REGLET) DETAIL

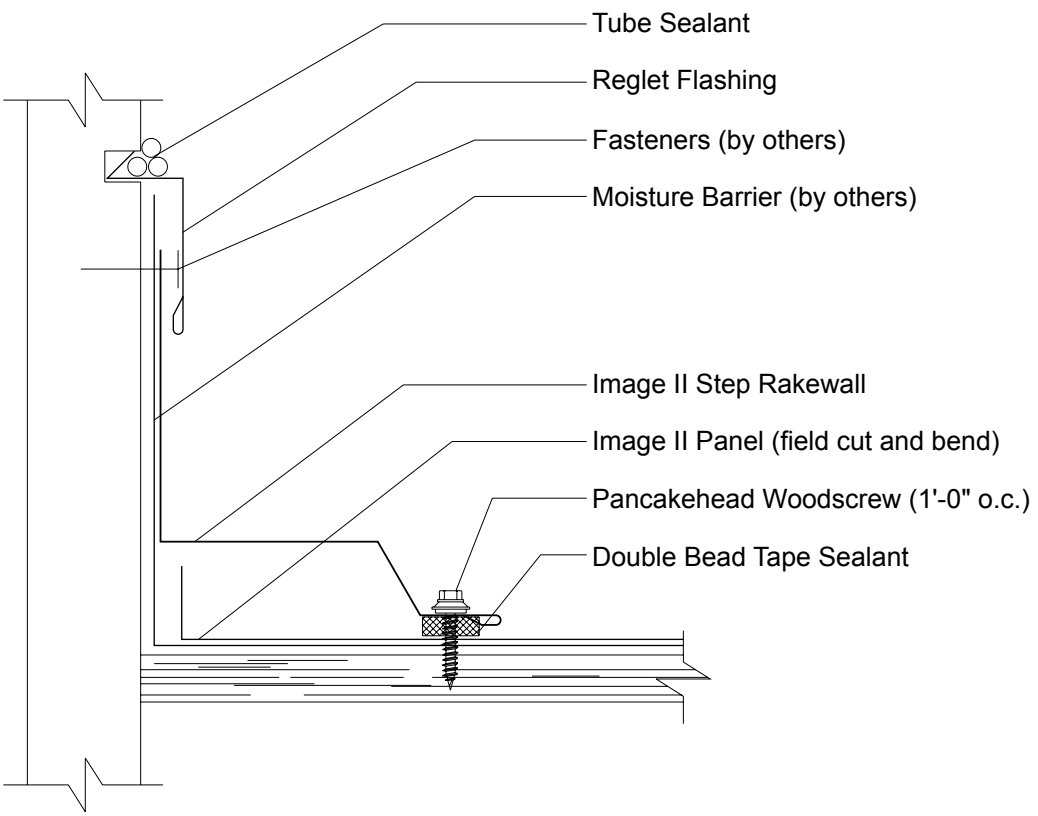
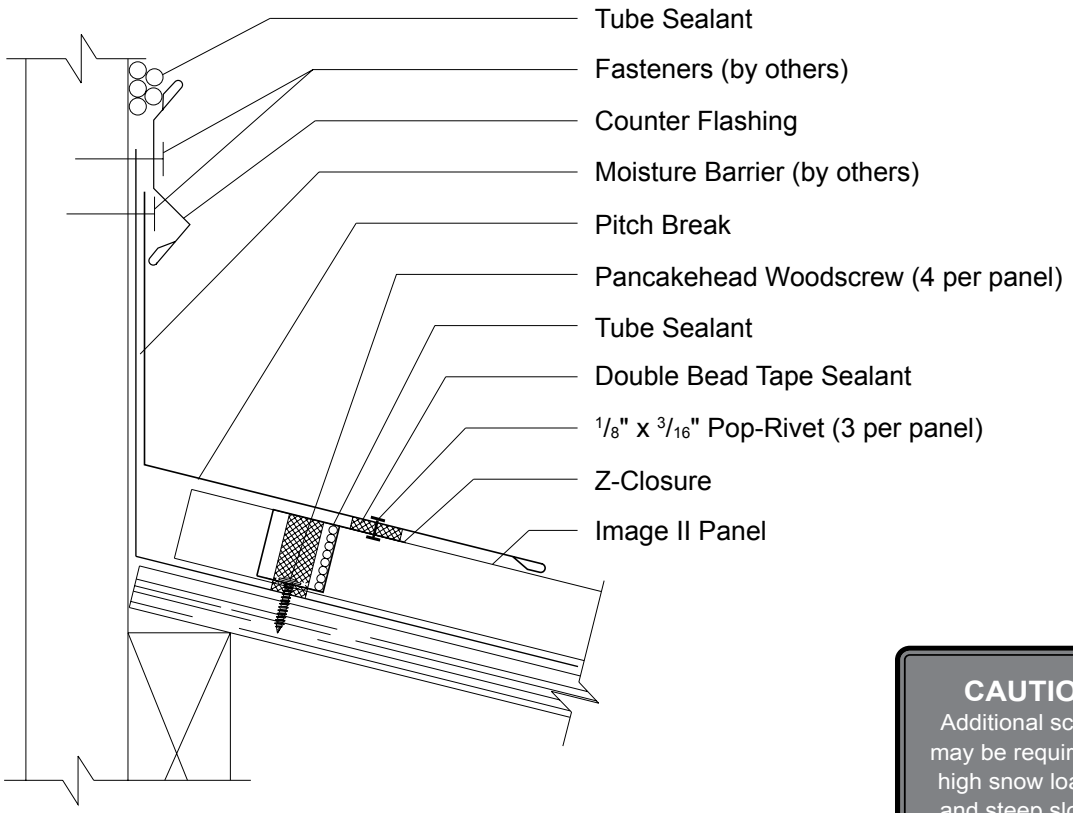


IMAGE II ENDWALL WITH Z-CLOSURE (COUNTER) DETAIL



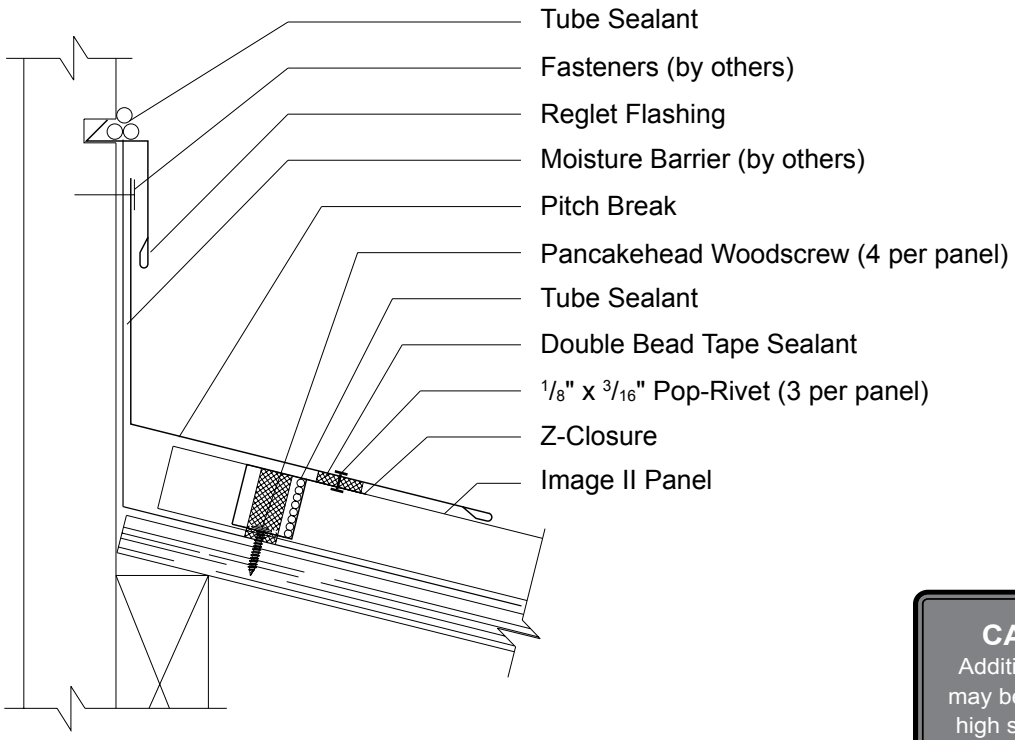
CAUTION
Additional screws may be required for high snow loading and steep slopes.

Note: Z-Closures must be field cut and bent to fit between panel ribs (see Architectural Manual).

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IMAGE II Endwall with Z-Closure (Reglet) Detail

3:12 Slope Minimum

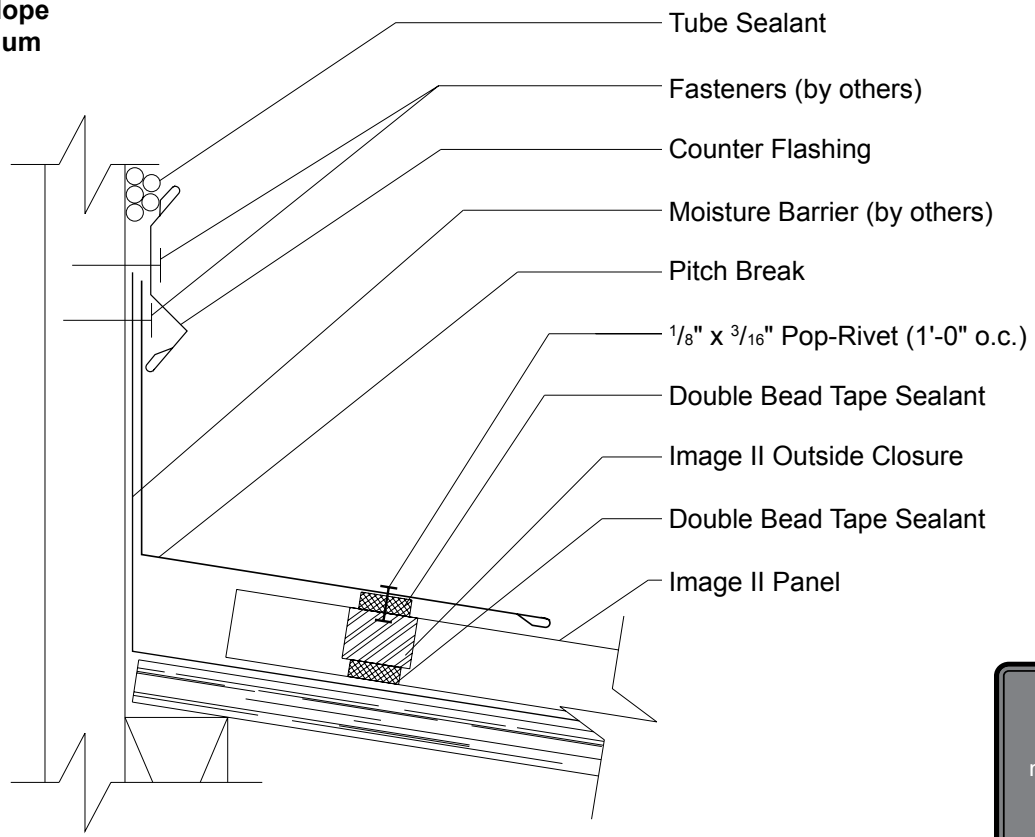


Note: Z-Closures must be field cut and bent to fit between panel ribs (see Architectural Manual).

CAUTION
Additional screws may be required for high snow loading and steep slopes.

IMAGE II Endwall with Foam-Closure (Counter) Detail

3:12 Slope Minimum

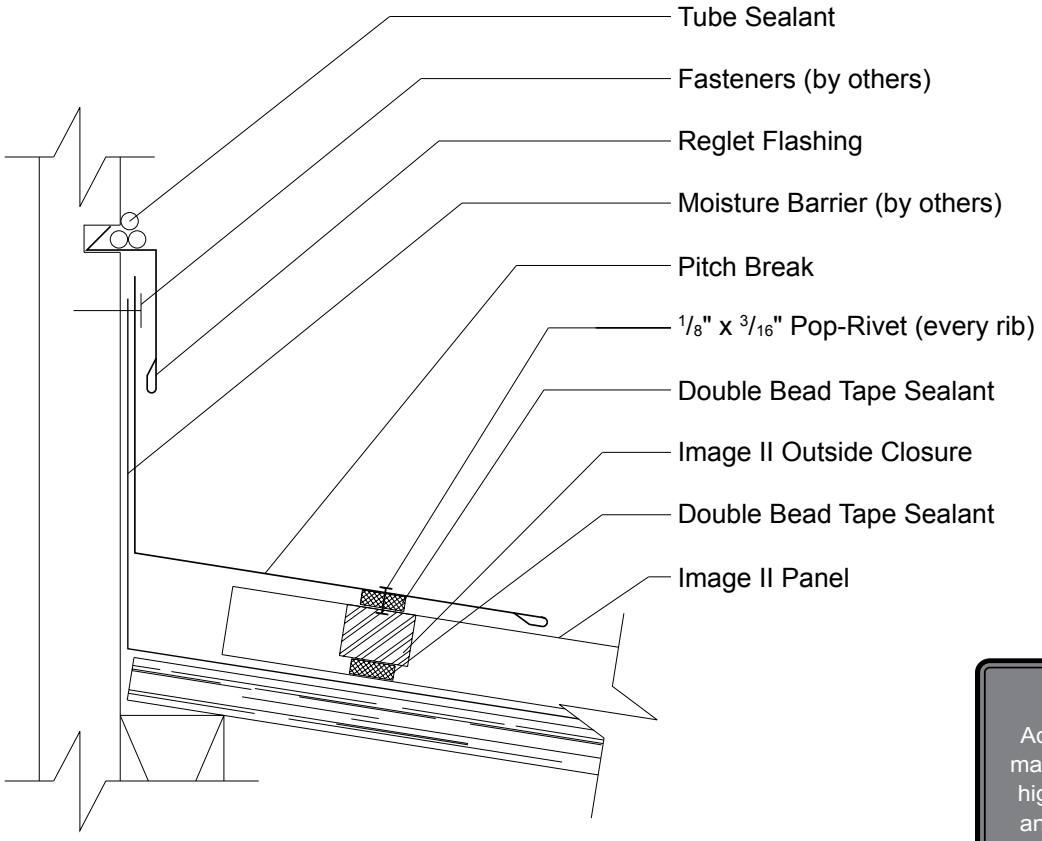


CAUTION
Additional screws may be required for high snow loading and steep slopes.

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IMAGE II Endwall with Foam-Closure (Reglet) Detail

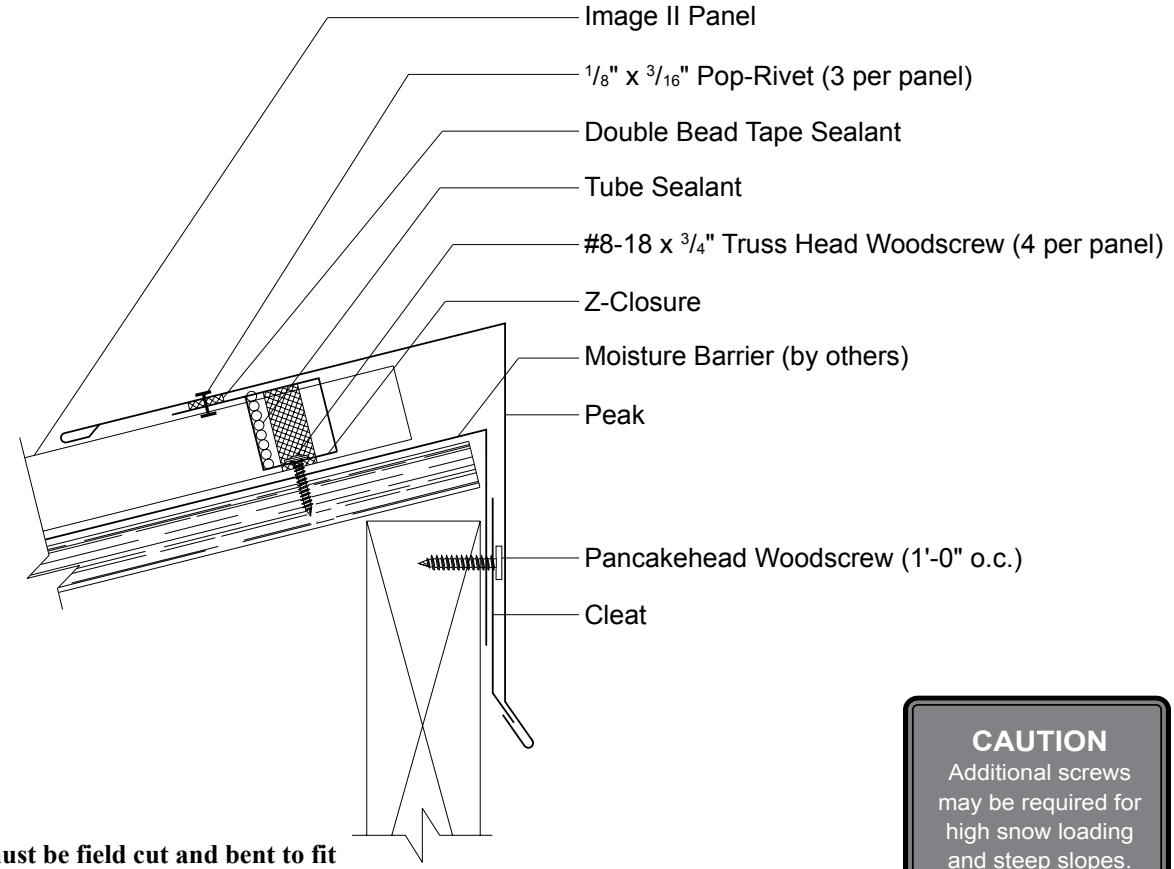
3:12 Slope Minimum



CAUTION
Additional screws may be required for high snow loading and steep slopes.

IMAGE II Peak with Z-Closure Detail

3:12 Slope Minimum



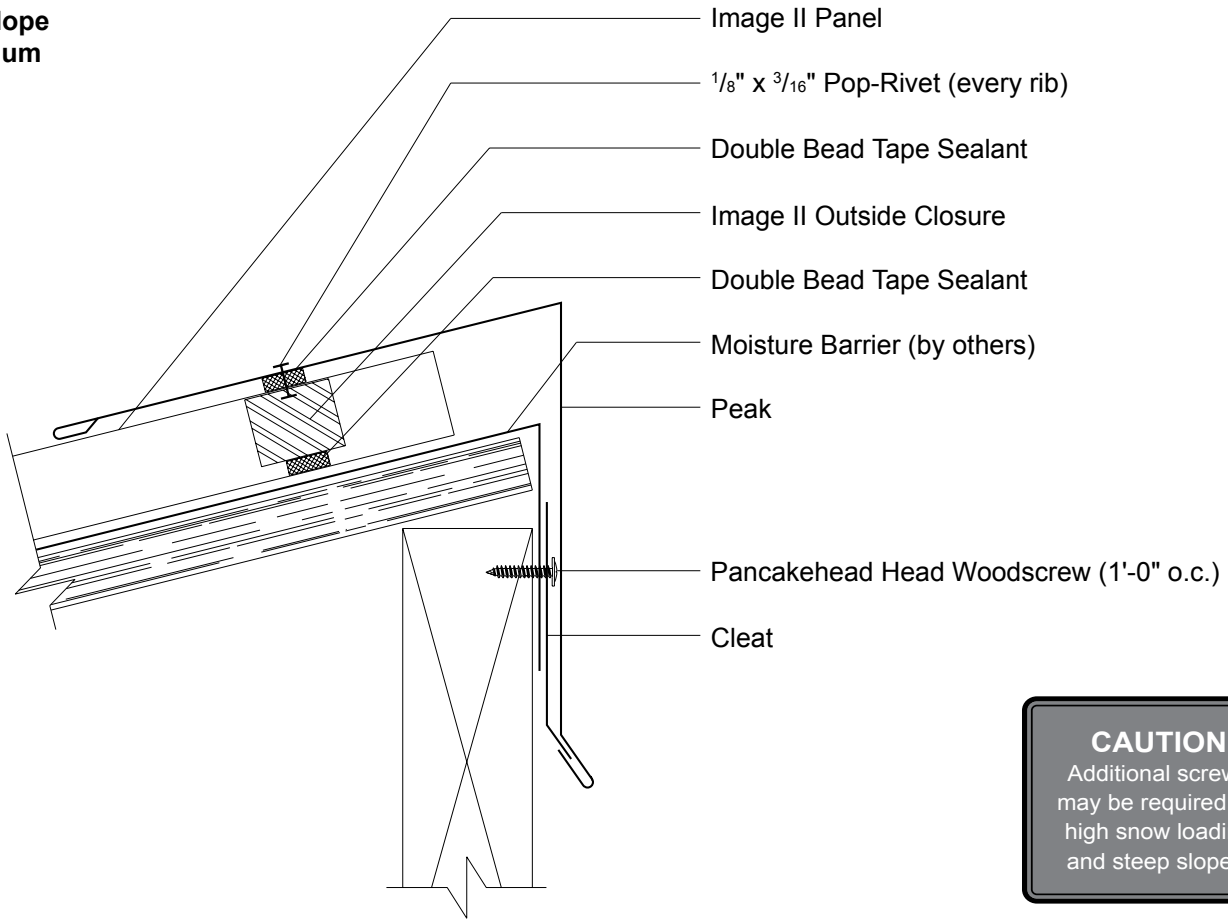
CAUTION
Additional screws may be required for high snow loading and steep slopes.

Note: Z-Closures must be field cut and bent to fit between panel ribs (see Architectural Manual).

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IMAGE II Peak with Foam-Closure Detail

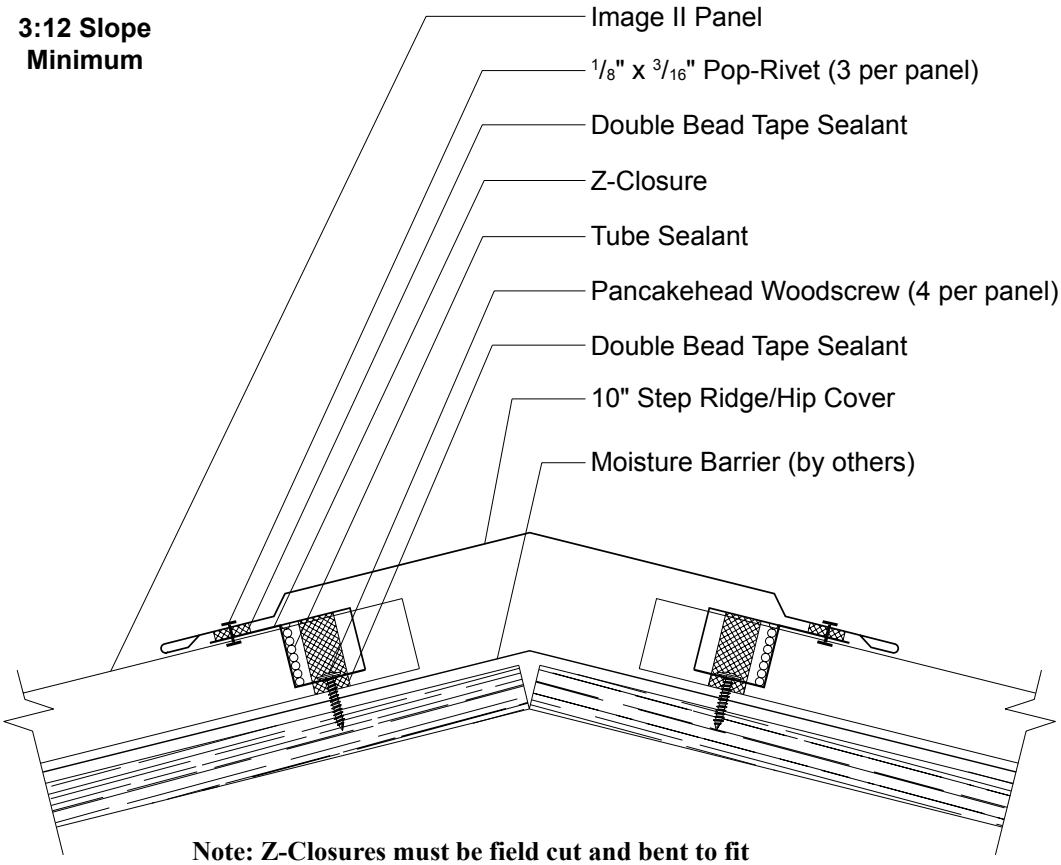
3:12 Slope Minimum



CAUTION
Additional screws may be required for high snow loading and steep slopes.

IMAGE II Ridge/Hip Cover Detail

3:12 Slope Minimum



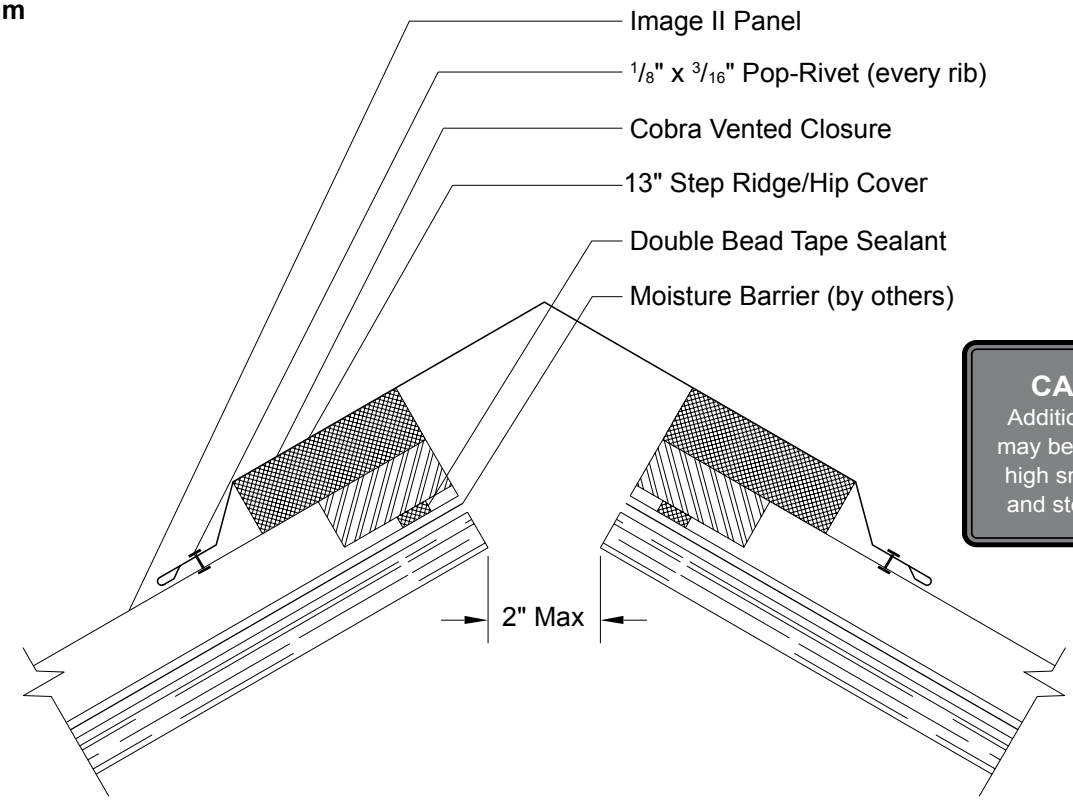
CAUTION
Additional screws may be required for high snow loading and steep slopes.

Note: Z-Closures must be field cut and bent to fit between panel ribs (see Architectural Manual).

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IMAGE II VENTED RIDGE (COBRA VENT) DETAIL

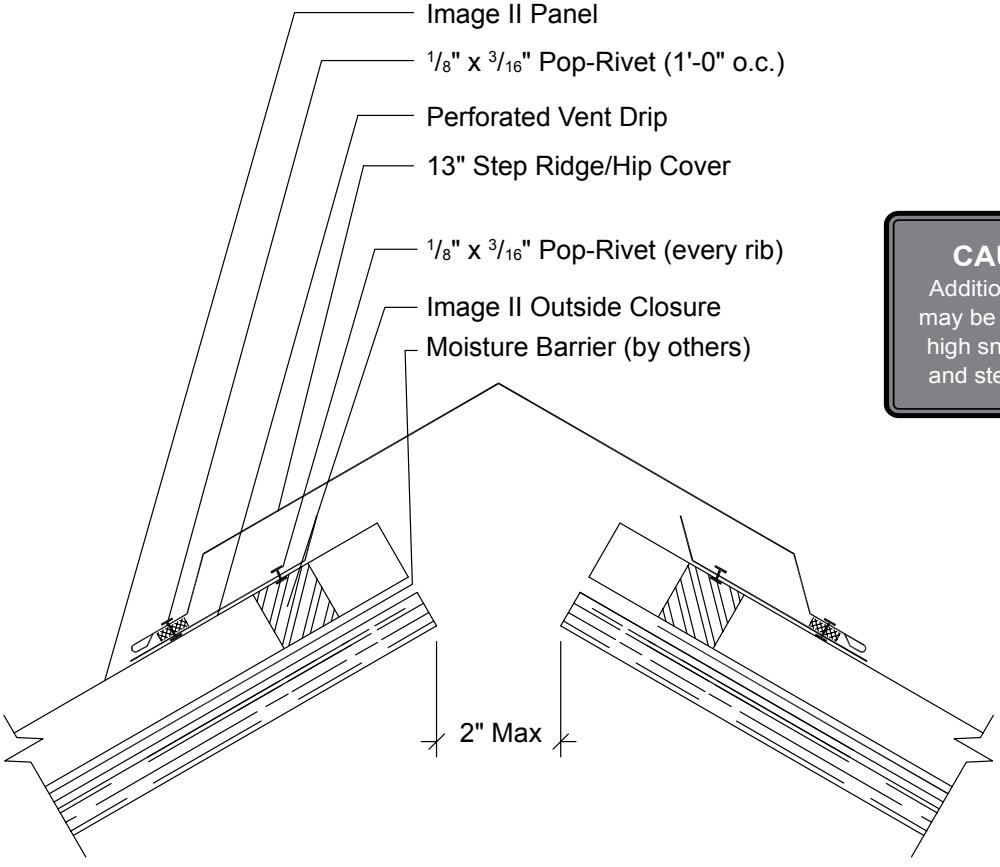
3:12 Slope Minimum



CAUTION
Additional screws may be required for high snow loading and steep slopes.

IMAGE II VENTED RIDGE (PERFORATED VENT DRIP) DETAIL

3:12 Slope Minimum



CAUTION
Additional screws may be required for high snow loading and steep slopes.

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